

ABHINAV REDDY CHERUKUPALLI

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PROFILE

Data Science student at the University of California, San Diego with interests in Machine Learning and Finance. Experienced in building predictive financial models and recommendation systems using Python. Proven researcher with a background in analyzing socioeconomic barriers in education using data-driven methodologies.

EDUCATION

University of California, San Diego (UCSD)

Bachelor of Science in Data Science, San Diego

- Bachelor of Science in Data Science | GPA: 3.54
- Relevant Coursework: DSC 10 (Principles of Data Science), DSC 20 (Algorithmic & Data Structures), MATH 18 (Linear Algebra), MATH 20B (Calculus for Science and Engineering), Cogs 9

TECHNICAL PROJECTS

Predictive Stock Market Analysis & Risk Assessment

Tools: Python, YFinance API, Matplotlib, NumPy

- Data Extraction:** Developed a financial data pipeline using the yfinance API to extract historical stock data for major tech equities.
- Statistical Analysis:** Engineered functions to calculate Daily Returns and Moving Averages to identify market volatility.
- Risk Modeling:** Performed Monte Carlo Simulations to forecast 10,000+ potential outcomes and quantify risk via Value at Risk (VaR).
- Visualization:** Created heatmaps and distribution plots using Seaborn to visualize stock correlations.

Movie Recommendation System (NSDC Project)

Tools: Python, Scikit-Learn, Pandas, SVD

- Algorithmic Implementation:** Built a collaborative filtering engine using Singular Value Decomposition (SVD) to predict user ratings.
- Performance Optimization:** Evaluated model accuracy using Root Mean Square Error (RMSE), iteratively tuning hyperparameters.
- Data Wrangling:** Processed large-scale movie datasets with pandas, handling sparse matrices for over 100k+ entries.

RESEARCH EXPERIENCE

Impact of Infrastructure & Socioeconomic Status on Rural Education

Independent Research | Primary Investigator

- Objective:** Investigated how physical infrastructure and socioeconomic status (SES) influence educational outcomes in rural Indian communities.
- Data Analysis:** Analyzed ASER data indicating that while 96.7% of children are enrolled in schools, basic infrastructure remains a barrier, with only 61.75% of schools having accessible toilets and 11% offering transportation.
- Key Findings:** Identified that distance to schools significantly impacts attendance, particularly for girls, and that lower SES correlates with higher dropout rates due to financial constraints.
- Technical Insights:** Utilized Python (Seaborn) to visualize the "Digital Divide," highlighting that while platforms like DIKSHA exist, fewer than 20% of rural schools have functional internet access.
- Actionable Recommendations:** Proposed a multi-faceted approach involving government infrastructure investment, community-based digital learning centers, and subsidized transportation.

TECHNICAL SKILLS

Python (Pandas, NumPy,	Jupyter Notebooks	Google Colab	Git/GitHub
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Scikit-Learn, Matplotlib, Seaborn) VS Code Monte Carlo Simulations	Linear Algebra Financial Modeling	Statistical Analysis	Data Cleaning
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AWARDS & LEADERSHIP

Distinction: World Economics Cup **2023**

A+ Rating: Global Economics Olympiad **2023**

Student Leader: Delhi Public School, Hyderabad **2022 to 2023**

Hyderabad

Coordinated activities for 150+ students and served as a liaison between students and administration.

HOBBIES

Basketball, Reading Books